2.5.24

EE25BTECH11058 - Mohana Krishna Sushma

Question:
If $|\mathbf{a}| = 2$, $|\mathbf{b}| = 7$ and $\mathbf{a} \times \mathbf{b} = 3\hat{\imath} + 2\hat{\jmath} + 6\hat{k}$, find the angle between $\mathbf{a}$ and $\mathbf{b}$.
Solution:
Let the vector $\mathbf{c} = \mathbf{a} \times \mathbf{b}$. We are given:

$$\|\mathbf{a}\| = 2 \tag{1}$$
$$\|\mathbf{b}\| = 7 \tag{2}$$
$$\mathbf{c} = \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \tag{3}$$

The squared magnitude of a vector $\mathbf{v}$ can be found using the matrix product $\mathbf{v}^\top \mathbf{v}$. We calculate the squared magnitude of $\mathbf{c}$:

$$\|\mathbf{c}\|^2 = \mathbf{c}^\top \mathbf{c} \tag{4}$$
$$= \begin{pmatrix} 3 & 2 & 6 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \tag{5}$$
$$= (3)(3) + (2)(2) + (6)(6) \tag{6}$$
$$= 9 + 4 + 36 \tag{7}$$
$$= 49 \tag{8}$$

Therefore, the magnitude of $\mathbf{c}$ is:

$$\|\mathbf{c}\| = \sqrt{49} = 7 \tag{9}$$

The magnitude of the cross product is also defined by the relation:

$$\|\mathbf{c}\| = \|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \, \|\mathbf{b}\| \sin(\theta) \tag{10}$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. Substituting the known values:

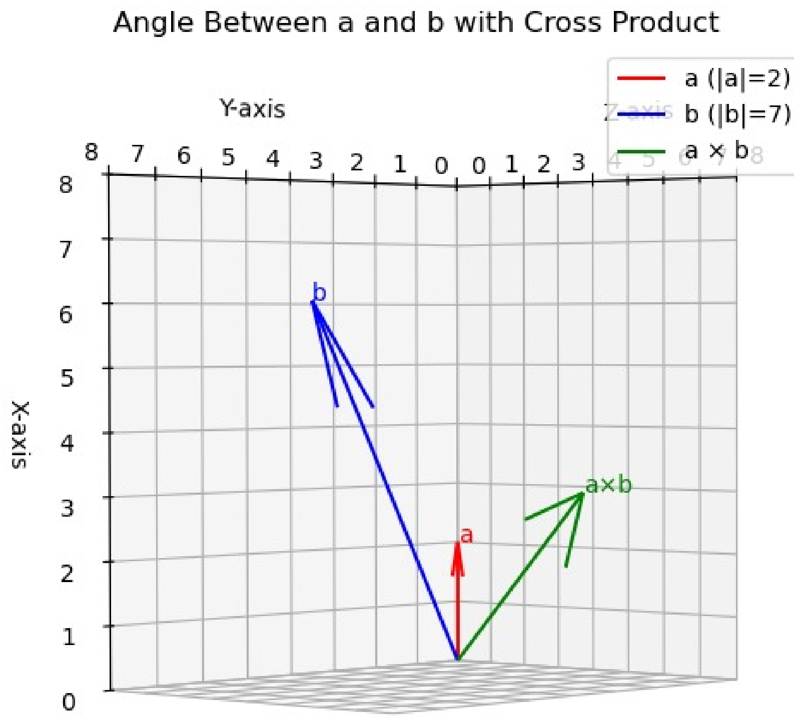
$$7 = (2)(7) \sin(\theta) \tag{11}$$
$$7 = 14 \sin(\theta) \tag{12}$$

Solving for $\sin(\theta)$:

$$\sin(\theta) = \frac{7}{14} = \frac{1}{2} \tag{13}$$

This implies that the angle θ is:

$$\theta = \arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6} \text{ or } 30^\circ \tag{14}$$


Angle Between a and b with Cross Product
a (|a|=2)
b (|b|=7)
a × b
Y-axis
Z-axis
X-axis
b
a
a×b